Ch 1.4

$AB = C$

$m \times n \quad n \times p \quad m \times p$

Dot prod → row i of A · col k of B

$i = 1 \to m \qquad k = 1$ to p

number $c_{ik} = m \times p$

col way → A times col k of B

$k = 1 \to p$

Col k of $C = p$ cols

row way → Row i of A times B

$i = 1$ to m

Row i of $C = m$ rows

col times rows → (col j of A)(row j of B)

$j = 1$ to n

Rank 1 Matrices

n-Mats

② $A = [a\ a\ a]$ $C = [a]$ $R = [1\ 1\ 1]$

⑤ 7 col 4 row · A 中 7 can't have ind cols more than rows as col vector belong to row dimension?

⑥ $$A = \begin{bmatrix} 2 & -2 & 1 & 6 & 0 \\ 1 & -1 & 0 & 2 & 0 \\ 3 & -3 & 0 & 6 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 3 & 0 & 1 \end{bmatrix}$$

⑦ $$R = \begin{bmatrix} 1 & -1 & 0 & 2 & 0 \\ 0 & 0 & 1 & \textcircled{2} & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

⑧ $$\begin{bmatrix} 3 & 4 \\ x & -3 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ x & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 9+4x & 0 \\ & \end{bmatrix}$$

$9+4x = 1$

$\therefore x = -\frac{8}{4} = -2$

(15) (i) col combⁿ of C

col j of A = Mat C times col j of R

(ii) Row combⁿ of R

Row i of A = row i of C times Mat R

(iii) (row i of C) . (col j of R) = A_{ij}

So col of C must = Row of R

(iv) r cols of C are indep.

r x r identity Mat ⊆ (so r indep rows)

(16) a) vector ABx = Mat A x vector Bx.

So combⁿ of cols of A.

so C(AB) ⊆ C(A)

b) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$

(a) (i) If A & B has rank 1

AB " " 1 or 0.

$A = uv^T$ & $B = xy^T$ $\therefore AB = u(v^Tx)y^T$

so AB = Zero Mat if dot prod of v^Tx is 0.

(b) A, B 3×3 mat of Rank 3, so AB has rank 3.

$\Rightarrow$ if $ABx = 0$ then $Bx = 0$ cz A has 3 indep cols

But $Bx = 0$ only if $x = 0$ cz b has 3 indep cols.

(c) Given, AB = BA for every 2×2 Mat. B.

$A = \begin{bmatrix} c & 0 \\ 0 & c \end{bmatrix} = cI$. only those mats

A = cI commute with every B.

$\Rightarrow$ choose $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ so, $AB = \begin{bmatrix} c & d \\ e & f \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} c & d \\ e & f \end{bmatrix}$

$\begin{bmatrix} c & d \\ e & 0 \end{bmatrix} = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix} \therefore d = e = 0$

Now, $B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $AB = \begin{bmatrix} c & 0 \\ 0 & f \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} c & 0 \\ 0 & f \end{bmatrix} \therefore \begin{bmatrix} 0 & c \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & f \\ 0 & 0 \end{bmatrix} \therefore c = f$

$A = cI$!

(18) $A = c =$ exchange Mat $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

(i) AB (row exchange) $= \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$

BC (Col exchange) $\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$

(ii) $(AB)C = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$ ↓ col exchange of AB

$A(BC) = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$ ↓ row Exchange of BC

Same

(10) col-row Multiplication:

$$A=\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad B=\begin{bmatrix} \text{ক} & \text{খ} & \text{গ} \\ \text{ঘ} & \text{ঙ} & \text{চ} \\ \text{ছ} & \text{জ} & \text{ঝ} \end{bmatrix}$$

$$AB=\begin{bmatrix} a \\ d \\ g \end{bmatrix}\begin{bmatrix} \text{ক} & \text{খ} & \text{গ} \end{bmatrix} + \begin{bmatrix} b \\ e \\ h \end{bmatrix}\begin{bmatrix} \text{ঘ} & \text{ঙ} & \text{চ} \end{bmatrix} + \begin{bmatrix} c \\ f \\ i \end{bmatrix}\begin{bmatrix} \text{ছ} & \text{জ} & \text{ঝ} \end{bmatrix}$$

$$BA=\begin{bmatrix} \text{ক} \\ \text{ঘ} \\ \text{ছ} \end{bmatrix}\begin{bmatrix} a & b & c \end{bmatrix} \ldots\ldots$$

(20) $AB=(4\times3)\times(3\times2)=4\times3\times2=24$

$(AB)C=(4\times2)\times(2\times1)=4\times2\times1=8$

Total $=24+8$

$=32.$